SAFETY MANUAL

Worcester Ball Valves:

44 Series

59 Series

459 Series

599 Series

Versions: Anti Static; FireSafe; Envirosafe; Cryogenic

Revision 0

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Experience In Motion

1. Introduction

This Safety Manual provides information necessary to design, install, verify and maintain a Safety Instrumented Function (SIF) utilizing Flowserve's Worcester Ball Valves. This manual provides necessary requirements for meeting the IEC 61508 or IEC 61511 functional safety standards.

1.1 Terms and Abbreviations

Safety:

Freedom from unacceptable risk of harm

Functional Safety:

The ability of a system to carry out the actions necessary to achieve or to maintain a defined safe state for the equipment / machinery / plant / apparatus under control of the system

Basic Safety:

The equipment must be designed and manufactured such that it protects against risk of damage to persons by electrical shock and other hazards and against resulting fire and explosion. The protection must be effective under all conditions of the nominal operation and under single fault condition

Safety Assessment:

The investigation to arrive at a judgment - based on evidence - of the safety achieved by safety-related systems

Fail-Safe State:

State where solenoid valve is de-energized and spring is extended

Fail Safe:

Failure that causes the valve to go to the defined fail-safe state without a demand from the process

Fail Dangerous:

Failure that does not respond to a demand from the process (i.e. being unable to go to the defined fail-safe state)

Fail Dangerous Undetected:

Failure that is dangerous and that is not being diagnosed by automatic stroke testing

Fail Dangerous Detected:

Failure that is dangerous but is detected by automatic stroke testing

Fail Annunciation Undetected

Failure that does not cause a false trip or prevent the safety function but does cause loss of an automatic diagnostic and is not detected by another diagnostic.

Fail Annunciation Detected:

Failure that does not cause a false trip or prevent the safety function but does cause loss of an automatic diagnostic or false diagnostic indication

Fail No Effect:

Failure of a component that is part of the safety function but that has no effect on the safety function

Low demand mode:

Mode, where the frequency of demands for operation made on a safety-related system is no greater than twice the proof test frequency

1.2 Acronyms

FMEDA:

Failure Modes, Effects and Diagnostic Analysis

MOC Management of Change:

These are specific procedures often done when performing any work activities in compliance with government regulatory authorities

PFD avg:

Average Probability of Failure on Demand

SFF Safe Failure Fraction:

The fraction of the overall failure rate of a device that results in either a safe fault or a diagnosed unsafe fault

***SIF Safety Instrumented Function*:**

A set of equipment intended to reduce the risk due to a specific hazard (a safety loop).

SIL Safety Integrity Level:

Discrete level (one out of a possible four) for specifying the safety integrity requirements of the safety functions to be allocated to the E/E/PE safety-related systems where Safety Integrity Level 4 has the highest level of safety integrity and Safety Integrity Level 1 has the lowest.

SIS Safety Instrumented System:

Implementation of one or more Safety Instrumented

1.3 Product Support

Please refer to the local Flowserve technical sales representative for support details.

2. Product Description

The Worcester Series 44 three-piece Ball Valve

The Flowserve Worcester Series 44 Ball Valve is the standard in the industry. Designed with advanced seal and seat technology, you are assured of maximum performance and life. Available through a nationwide network of distributors, you are also assured of wide availability and fast response. Series 44 can be used in the following applications: OEM, Petroleum Production and Refining, Synthetic Fuels, General Processing Industry, Utility, Commercial, Hydrocarbon Processing, Petrochemical Processing, Boiler Circulation, Chemical Transfer, Industrial Plant Services, Flammable Liquids, Water Treatment, Heat Transfer Fluids

Features

- 3Pc swing out design, tight shut off and bi-directional sealing
- High performance stem seal and seat design and options
- Designed for automation
- Multiple end connections

The Worcester Series 59 three-piece Ball Valve

The Flowserve Worcester Series 59 Ball Valve offers a full port option to the standard Series 44 Ball Valve. Full port capability is advantageous in applications where fluid velocity is a concern, such as slurries, viscous fluids and fluids with particulates. Designed to ASME B16.34 ratings and with an advanced seal technology, you are assured of maximum performance and life. Series 59 can be used in a wide range of processing applications.

Features

- 3pc swing out design, tight shut off and bi-directional sealing
- High performance stem seal and seat design and options
- Designed for automation
- Multiple end connections

The Worcester Series 459 three-piece Ball Valve

The Flowserve Worcester A459 Series ball valves, complements the Series 44 to provide the most versatile, reliable and widely available three piece range. With a wide range of seat materials available, the valve is suitable for a most applications within the process industry up to 50 Bar. Designed with an ISO 5211 mounting platform the valve is also fitted with a direction indicator that remains on the valve during actuation to maintain stem seal integrity and provide visual indication of the ball position.

Features

- ISO 5211 mounting platform
- Three piece design
- Position indicator remains during actuation
- Wide range of seat materials
- Mechanical anti-static device
- Locking clip of valve stem nut

The Worcester Series 599 three-piece Ball Valve

The Flowserve Worcester A599 Series ball valves, offers a full port option to the standard Series 459, to provide the most versatile, reliable and widely available three piece range. With a wide range of seat materials available, the valve is suitable for a most applications within the process industry up to 50 Bar. Designed with an ISO 5211 mounting platform the valve is also fitted with a direction indicator that remains on the valve during actuation to maintain stem seal integrity and provide visual indication of the ball position.

Features

- ISO 5211 mounting platform
- Three piece design
- Position indicator remains during actuation
- Wide range of seat materials
- Mechanical anti-static device
- Locking clip of valve stem nut

3.0 Designing a Safety Instrumented Function using a Worcester Ball Valve

Although Flowserve can, and often does, provide general guidelines, it is obviously not possible to provide application specific data and warnings for all conceivable applications. The purchaser/end-user must therefore assume the ultimate responsibility for the proper selection, installation, operation and maintenance of the products.
Read the appropriate IOM (User Instructions) supplied with the product, or obtain a copy from the local Flowserve representative, before installing, operating or repairing any valve. The purchaser/end user should fully train its employees and/or contractors in the safe handling and use of Worcester products in connection with the purchaser's manufacturing processes.

3.1 Safety Function

When de-energized, an automated ball valve moves to its fail-safe position. Depending on the actuator mounting specification of fail-open or fail-closed operation, the valve ball will either close off the flow path through the valve body or open the flow path through the valve body when de-energized.

The Worcester Ball Valves are intended to be part of final element subsystem as defined per IEC 61508 and the achieved SIL level of the designed function must be verified by the designer.

Seat Design
44, 459, 59 and 599 series ball valves feature Cavity Pressure Relieving (CPR) seats ensure that pressure generated through media expansion when the ball valve is closed is safely relieved upstream.

Ball Design
The ball utilises a pressure equalising hole to balance cavity pressure with line pressure, when the valve is open.

Stem Design
The ball valves feature an anti-blowout stem, inserted from inside the body, for greater safety

3.2 Environmental limits

The designer of a SIF must check that the product is rated for use within the expected environmental limits. Refer to the Flowserve/Worcester Ball Valve Brochure for environmental limits, or refer to Flowserve technical support.

3.3 Application limits

The materials of construction of a Worcester Ball Valve are specified in the appropriate brochure. It is especially important that the designer checks for material compatibility considering the on-site chemical contaminants and air supply conditions. If the Valve is used outside of the application limits or with incompatible materials, the reliability data provided becomes invalid.

3.4 Design Verification

A detailed Failure Mode, Effects, and Diagnostics Analysis (FMEDA) report is available from Flowserve. This report details all failure rates and failure modes as well as the expected lifetime.

The achieved Safety Integrity Level (SIL) of an entire Safety Instrumented Function (SIF) design must be verified by the designer via a calculation of PFD avg considering architecture, proof test interval, proof test effectiveness, any automatic diagnostics, average repair time and the specific failure rates of all products included in the SIF. Each subsystem must be checked to assure compliance with minimum hardware fault tolerance (HFT) requirements.

The failure rate data listed the FMEDA report is only valid for the useful life time of the valve. The failure rates will increase sometime after this time period. Reliability calculations based on the data listed in the FMEDA report for mission times beyond the lifetime may yield results that are too optimistic, i.e. the calculated Safety Integrity Level will not be achieved.

3.5 SIL Capability

3.5.1 Systematic Integrity

The product has met manufacturer design process requirements of Safety Integrity Level (SIL) 3. These are intended to achieve sufficient integrity against systematic errors of design by the manufacturer. A Safety Instrumented Function (SIF) designed with this product must not be used at a SIL level higher than the statement without “prior use” justification by end user or diverse technology redundancy in the design.

3.5.2 Random Integrity

The Worcester Ball Valve is a Type A Device (‘non-complex’ subsystem using discrete elements). Therefore based on the SFF between 60% and 90%, when the valve is used as the only component in a final element subassembly, a design can meet SIL 2 at HFT=0.

When the final element assembly consists of many components (Ball Valve, actuator, etc.) the SIL must be verified for the entire assembly using failure rates from all components. This analysis must account for any hardware fault tolerance and architecture constraints.

3.5.3 Safety Parameters

For detailed failure rate information refer to the Failure Modes, Effects and Diagnostic Analysis Report for the Worcester Ball valve.

3.6 General Requirements

The system’s response time shall be less than process safety time as defined in the SIF Safety Requirements Specification. Since the response time of the Worcester Ball Valve depends upon the mounted actuator and associated devices, the response time must be confirmed for each final element assembly. All SIS components including the Worcester Ball Valve must be operational before process start-up. Speed of closure (nominal speed and base conditions) is dependent upon valve actuation and end-user installation.

User shall verify that the Worcester Ball Valve is suitable for use in safety applications by confirming the Worcester Ball Valve’s nameplate is properly marked. Any personnel performing maintenance, repairs and testing on the Worcester Ball Valve must be competent to do so.
Refer to the DFMEA report for information regarding the useful life of the Worcester Ball Valve.

4. Installation and Commissioning

4.1 Installation

Read the appropriate Worcester Ball Valve IOM, as supplied with the valve, before installing, operating or repairing any valve. The purchasers/end-user should fully train its employees and/or contractors in the safe use of Worcester products in connection with the purchaser’s manufacturing processes. The physical environment must be checked to verify that conditions do not exceed the ratings.

4.2 Physical Location and Placement

Read the appropriate Worcester Ball Valve IOM, as supplied with the valve, before installing, operating or repairing any valve. The Worcester Ball Valve should be accessible with sufficient room for physical inspection and manual proof testing. With regards to the valve actuating, allow for a sufficient dimensional envelope for connections and supply piping.

5. Operation and Maintenance

5.1 Proof test without automatic testing

Refer to applicable FMEDA for proof testing information.

The objective of proof testing is to reveal potentially dangerous faults within the Valve that are not detected by any automatic diagnostics of the system. Of main concern are undetected failures that prevent the safety instrumented function from performing its intended function.

The frequency of proof testing, or the proof test interval, is to be determined in reliability calculations for the safety instrumented functions. The proof tests must be performed more frequently than or as frequently as specified in the calculation in order to maintain the required safety integrity of the safety instrumented function.

The following proof test is recommended. The results of the proof test should be recorded and any failures that are detected and that compromise functional safety should be reported to Flowserve immediately.
The suggested proof test consists of a full stroke of the valve and valve actuator– Refer to Table 16, Appendix B of the relevant FMEDA report. The person(s) performing the proof test of a Valve should be trained in SIS operations, valve maintenance and company Management of Change procedures.

5.3 Repair and replacement

Repair procedures for the Worcester Ball Valve are located in the IOM, as supplied with the valve, or a copy can be obtained from the regional Worcester office.

5.4 Useful Life

Based on general data, proper use and correct maintenance, the useful life of the Worcester Ball Valve is on *average* 10-15 years. Refurbishment will extend the product life.

5.5 Flowserve Notification

Any failures that are detected and that compromise functional safety should be reported to Flowserve immediately.

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To find your local Flowserve representative please use the Sales Support Locator System found at www.flowserve.com

Or call toll free: 1 800 225 6989

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